VF-B07 Trouser Tear

Test Method for Trouser Tear	Test method #: VF- B07 REF #: TM-R -042 & TM-R- 050
Date Issued: November 18, 2022 Revision #: 6	Reason for Revision: 1. Update references 2.Rectify thickness gauge capable of measuring to 0.01mm

PURPOSE AND SCOPE: To measure the load required to tear a material along a line cut perpendicular to its thickness.

REFERENCES: ASTM D2261, ASTM D4704, ASTM D624

EQUIPMENT AND MATERIAL:

1. Tensile testing machine, meeting the requirements of ASTM D638
2. Clamping jaws, preferable rubber faced.
3. Thickness gauge, capable of measuring to 0.01mm
4. Vernier caliper, capable of measuring to 0.01mm
5. Leather, synthetic leather, fabric, rubber, box, plastic.

SAMPLE PREPARATION: 1

1. Condition the sample for at least 24 hours at the standard laboratory conditions at 23.0±2.0˚C Temperature and 55 ±10 % Relative Humidity.

2. Die or hand cut samples according to the requirements of the specific material type:

A. *Fabric:*

1. The test samples should be taken in both the warp and the fill (Weft) direction. Samples should be taken such that no two specimens contain the same warp of fill yarns. Samples should always be taken from different areas of the fabric and at least 10cm from any edge.
2. Cut specimens with their long dimensions parallel to the warp (machine) direction and specimens with their long dimensions parallel to the filling (cross) direction. Label the machine direction as direction “A” or “L” and the filling direction as direction “B” or “W”

B. *Leather:*

Test samples should be evaluated in both the length **(L)** and width **(W)** directions. **For leather**, the length direction is parallel to the backbone. **For non-woven backed synthetic leather,** the length direction corresponds to the machine direction.

REPETITIONS:

Test three (3) samples per material. For materials which are direction dependent, test three materials in each direction.

VF-B07 Trouser Tear

TEST PROCEDURE:

Test should be carried out at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. When evaluating materials of unspecified thickness (typically rubber and plastic), measure the thickness (see VF-A03) of the materials at the initial tearing location and two other points along the “tearing line”. Record the thinnest of the measurements to the nearest 0.1mm **(T)** **Note**: If thickness readings vary by more than +/-0.2mm, the deviating samples should either be rebuffed or disregarded and replaced.

2. Zero and calibration the tensile testing machine.
3. Set the tensile testing machine at the specified crosshead speed and Jaw gap (see *sections A-C, within Requirements for specific materials*).
4. Place the test sample in the jaws such that the slit is centered in the jaws and one of the “legs” is held in each of the jaws. There should be no tension on the sample, but it must be taught in the jaws (no slack) and equal portions of the test material should be positioned in the top and bottom jaws.
5. Operate the tensile testing machine at the specified crosshead speed *(see sections A-C, within Requirements for specific materials).*
6. Record the tearing load (L) according to the requirements of each material type *(see sections AC, on the requirements for specific materials).*

CALCULATIONS:

A. Materials of specified thickness
(These materials typically include leather, synthetic leather, and fabric).

Trouser tear strength (L) is simply reported in kgf.

B. Materials of unspecified thickness
(These materials typically include rubber, plastic, and foam).

Trouser tear strength (kgf /cm) = L/T

REPORT: The average of three tears strength measurements to the nearest 0.1kgf or 0.1kgf/cm.

Requirements for Specific Materials:

A). Leather, woven, & non-woven-backed synthetic leather (ASTM D4704).

1. Testing Parameters

Sample size = see **Figure A**

Crosshead speed = 250mm/min

Jaw gap =50 ± 5mm

Sample thickness = as is

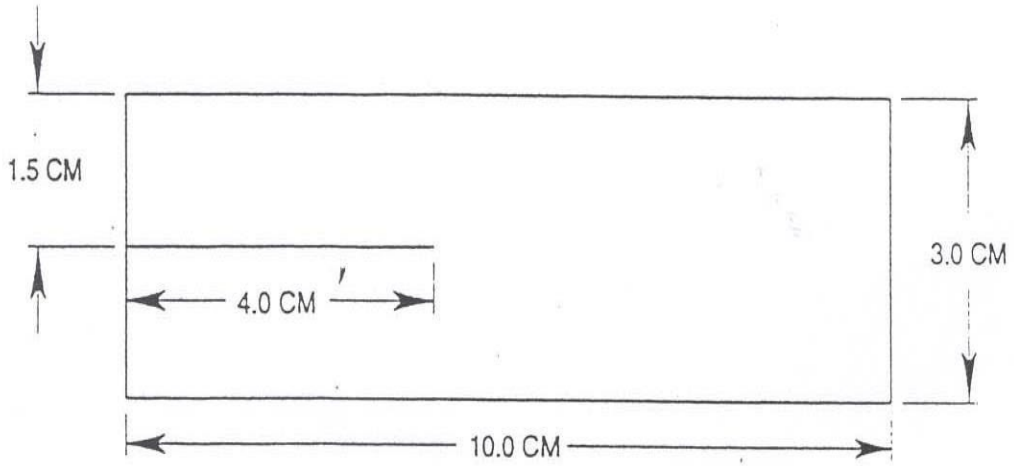


Figure A

B). Fabric (ASTM D2261)

1. Testing Parameters

Sample size = see **Figure B**

Crosshead speed = 300mm/min

Jaw gap = 75 ± 5mm

Sample thickness = as is

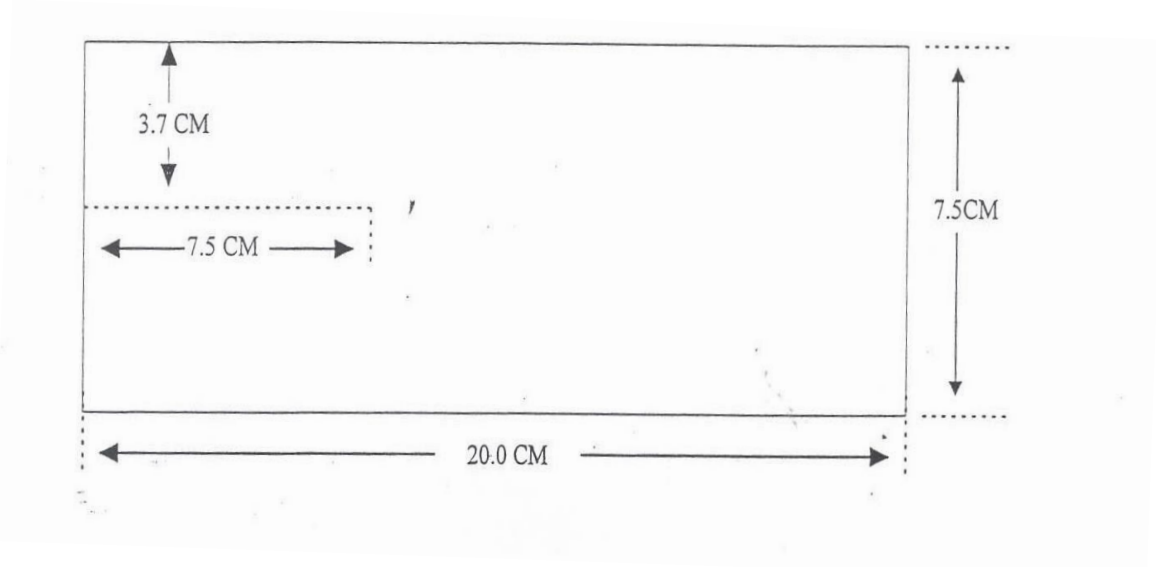


Figure B

1. Measurement of Tear Strength

The tearing strength is measured as the average of the five (5) highest peak loads (Please see figure C). The test should be conducted until the fabric has been torn a minimum of 50mm and until 5 peaks have been obtained. Report the load in kgf ,to the nearest 0.1kgf . The length direction corresponds to the slit being perpendicular to the warp yarns.

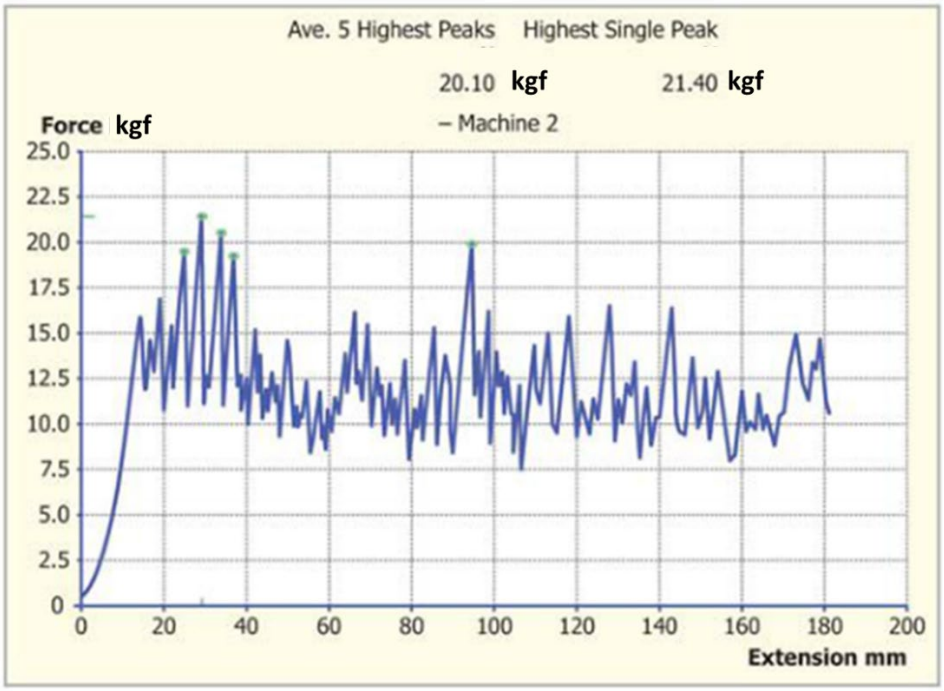


Figure C

C). Rubber, Plastics, Box (ASTM D624)

1. Testing Parameters

Sample size = see **Figure D**

Crosshead speed = 100mm/min

Jaw gap =40 ± 4mm

Sample thickness = 2.5 ± 0.5mm

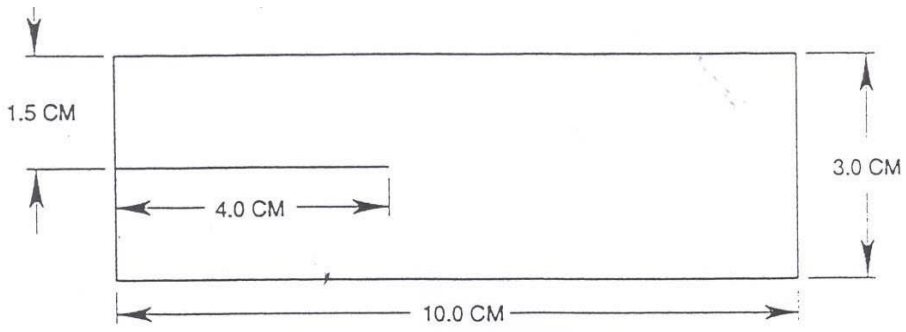


Figure D

2. Measurement of Tearing Strength

For plastics and Rubber, the tearing strength is measured as the average of the peak loads over the entire tearing length. Report the load in kgf/cm to the nearest 0.1kgf/cm.

For boxes, the sample must be prepared by placing a strip of tape parallel to and on either side of the 4.0cm long slice in the material created by the die. The tearing strength is measured as the load at the instant of the initial tear. The test should be conducted until the sample has been torn for the complete 60mm. The length direction corresponds to the slit being perpendicular to the machine direction. Report the load in kgf to the nearest 0.1kgf.